**App**

+ CELLSIZE: int
+ SIDEBAR: int
+ TOPBAR: int
+ BOARD_WIDTH: int
+ background_R: int
+ background_G: int
+ background_B: int
+ WIDTH: int
+ HEIGHT: int
+ FPS: int
+ configPath: String
+ random: Random
+ mapTile: PImage[][]
+ Map_Grid: char[][]
- wizard_house_display_x: int
- wizard_house_display_y: int
+ wizard_house_x: int
+ wizard_house_y: int
- wizardImage: PImage
- topbar: Topbar
- sidebar: Sidebar
- sidebar: Sidebar
+ shortest_Paths: HashMap<MyPoint, ArrayList<MyPoint>>
+ startPoints: ArrayList<MyPoint>
+ FF_press_condition: boolean
+ Pause_press_condition: boolean
+ Build_tower_condition: boolean
+ Place_tower_on_map: boolean
+ Upgrade_mana_pool_condition: boolean
+ Upgrade_range_condition: boolean
+ Upgrade_speed_condition: boolean
+ Upgrade_damage_condition: boolean
+ Upgrade_damage_condition: boolean
+ Deploy_range_upgrade: boolean
+ Deploy_speed_upgrade: boolean
+ Deploy_damage_upgrade: boolean
+ Lose_condition: boolean
+ mouse_on_which_button: String
+ initial_mana: int
+ initial_mana_cap: int
+ initial_mana_gained_per_second: int
+ current_mana_value: int
+ mana_pool_spell_initial_cost: int
+ mana_pool_spell_cost_increase_per_use: int
+ mana_pool_spell_cap_multiplier: double
+ mana_pool_spell_mana_gained_multiplier: double
+ initial_tower_range: double
+ initial_tower_firing_speed: double
+ initial_tower_damage: int
+ initial_tower_cost: int
+ waves: JSONArray
- gremlin_image: PImage
# tower_0_image: PImage
# tower_1_image: PImage
# tower_2_image: PImage
# bullet_image: PImage
- tower_image_map: HashMap<Integer, PImage>
- next_wave_pause: double
+ wave_duration: double
+ pre_wave_pause: double
+ gremlinList: ArrayList<Gremlin>
+ death_gremlin_index: ArrayList<Gremlin>
- monster_info_map: HashMap<Integer, ArrayList<Object>>
+ gremlin_dead: PImage[]
+ monster_type: String
+ monster_hp: double
+ monster_speed: double
+ monster_armour: double
+ mana_gained_on_kill: int
+ monster_quantity: double
+ monster_count: double
- Monster_timer: double
- Monster_timer: double
- spawn_point: MyPoint
- startTime: int
- startsIn: int
- startsIn: int
- current_wave: int
- next_wave_index: int
- next_wave_index: int
+ next_wave_display: int
- processWave: boolean
- isLastWave: boolean
- lastWaveFinished: boolean

+ App()
+ getStartIn(): int
+ readLevel(String levelAddress): char[][]
+ choosePath(int x, int y, char[][] backGroundArray): int
+ loadBackGround(char[][] backGroundArray): void
+ settings(): void
- findShortestPath(HashMap<MyPoint, MyPoint> pair, MyPoint start, MyPoint destination): void
- BreadthFirstSearch(MyPoint start, MyPoint destination): ArrayList<MyPoint>
+ setup(): void
+ keyPressed(): void
+ keyReleased(): void
+ mousePressed(MouseEvent e): void
+ mouseReleased(MouseEvent e): void
+ wave_change(): void
+ process_wave(int cur_wave): void
+ monster_generate(): void
+ draw(): void
+ rotateImageByDegrees(PImage pimg, double angle): PImage
+ method: void

**Topbar**

- mana_width: int
- mana_height: int
- mana_X: int
- text_y: int
- mana_inner_width: int
- mana_box_x: int
- mana_box_y: int
- mana_info_x: int
- mana_counter: int

+ Topbar()
+ mana_counter_sub(int sub_value): void
+ draw(App app): void

**Sidebar**

- sideBar_button_x: int
- sideBar_button_y: int
- rangeCost: int
- speedCost: int
- damageCost: int
- manaPoolCost: int
+ ffButton: Button
+ pauseButton: Button
+ buildTowerButton: Button
+ upgradeRangeButton: Button
+ upgradeSpeedButton: Button
+ upgradeDamageButton: Button
+ manaPoolButton: Button
- tower_list: ArrayList<Tower>
- tower_to_point: HashMap<Tower, MyPoint>
+ point_to_tower: HashMap<MyPoint, Tower>
- tower: Tower
- costs_board: ArrayList<String>
+ total_cost: int

+ Sidebar()
+ draw(App app): void

**Button**

- button_cost: int
- content_in_rect: String
- content_alongSide: String
- button_x: int
- button_y: int
- button_width: int
- button_hight: int

+ Button(int button_cost, String content_in_rect, String content_alongSide, int button_x, int button_y)
+ getX(): int
+ getY(): int
+ set_button_cost_and_content(int cost, String content): void
+ mouse_on_button(App app): boolean
+ drawCost(App app): void
+ drawCostValue(App app, int value): void
+ drawGrey(App app): void
+ drawYellow(App app): void
+ drawText(App app): void
+ drawMouse(App app): void
+ draw(App app): void

**MyPoint**

+ x: int
+ y: int

+ MyPoint()
+ MyPoint(int x, int y)
+ setXY(int x, int y): void
+ getX(): int
+ getY(): int
+ equals(Object obj): boolean
+ toString(): String
+ hashCode(): int

***Monster***

+ monster_x: int
+ monster_y: int
+ speed: double
+ sprite: PImage
+ isAlive: boolean
+ isReachTheWizardhouse: boolean
+ hp: double
+ armour: double

+ Monster(int monster_x, int monster_y, double speed, double hp, double armour, boolean isAlive, boolean isReachTheWizardhouse, PImage sprite)
+ setMonsterImage(PImage sprite): void
+ getX(): int
+ getY(): int
*+ tick(): void*
*+ draw(App app): void*

**Gremlin**

+ animation_start: boolean
+ animation_finished: boolean
+ current_frame: int
+ number_frames: int
- life_bar_width: int
- next_point: MyPoint
- indexMap: HashMap<MyPoint, Integer>
- current_speed: double

+ Gremlin(int monster_x, int monster_y, double speed, double hp, double armour, boolean isAlive, boolean isReachTheWizardhouse, boolean animation_start,PImage sprite)
+ set_gremlin_XY (int x, int y): void
+ drawGremlinDeath(int x, int y, App app): void
+ setSpawnPosition(int x, int y): void
+ resume_speed(): void
+ FF_speed(): void
+ hitted(double damage_take): void
+ tick(): void
+ get_sprite_width(): int
+ get_sprite_height(): int
+ draw(App app): void

**Tower**

- tower_x: int
- tower_y: int
- tower_range: double
- tower_firing_speed: double
- tower_damage: double
- range_level: int
- speed_level: int
- damage_level: int
- upgrade_count: int
- tower_cost: int
- range_cost: int
- speed_cost: int
- damage_cost: int
- tower_sprite: PImage
- bullet: Bullet

+ Tower(double initial_tower_range, double initial_tower_firing_speed, int initial_tower_damage, int initial_tower_cost)
+ mana_subtractor(int subtract_Cost): void
+ set_tower_cost(int cost): void
+ set_tower_cost(int cost): void
+ get_tower_cost(): int
+ get_X(): int
+ get_Y(): int
+ setTowerImage(PImage sprite): void
+ setTowerXY(int tower_x, int tower_y): void
+ setTowerWithMouse(App app): void
+ upgrade_cost(): void
+ upgrade_range(): void
+ getRangeLevel(): void
+ getRangeCost(): void
+ upgrade_speed(): void
+ getSpeedLevel(): void
+ getSpeedCost(): void
+ upgrade_damage(): void
+ getDamageLevel(): void
+ getDamageCost(): void
+ draw_O(App app, int level): void
+ draw_X(App app, int level): void
+ draw_square(App app, int level): void
+ draw_range(App app): void
+ mouse_on_tower(App app): boolean
+ valid_place_tower(App app): boolean
+ getClosestMonsterInRange(List<Gremlin> gremlinList): Gremlin
+ set_bullet(): void
+ fire_bullet(App app): void
+ drawUpgradePanel(int x, int y, int width, int height, ArrayList<String> costs, App app): void
+ draw(App app): void

***<<interface>>***
***FireBullet***

+ set_bullet(): void
+ fire_bullet(App app): void

**Bullet**

- bullet_x: int
- bullet_y: int
- move_distance: int
- sprite: PImage
- bullet_damage: double
- return_time: double

+ Bullet (int bullet_x, int bullet_y, double bullet_damage, PImage sprite)
+ getX(): int
+ getY(): int
+ getDamage(): double
+ setBulletXY(int x, int y): void
+ update_damage(double damage): void
+ moveToMonster(Gremlin gremlin, int tower_x, int tower_y): void
+ draw(App app): void